Gonzalo Cruz Gómez

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Data Science & Engineering undergraduate (URJC, graduating June 2027) with a background in Telecommunications Engineering. Builds data systems end to end: orchestration, streaming, and models that hold up on real data. Currently on Repsol's Data Intelligence Platform team; founder of Asociación Datalab.

Education

B.S. Data Science & Engineering **URJC** *2024 – 2027*

- Final year (expected graduation June 2027). Completed two academic years in one (2024–2025). Coursework: ML, Distributed Systems, Deep Learning, HPC, Statistical Inference.

B.S. Telecommunications Engineering **UC3M** *2019 – 2023*

- 120+ credits in Networking, Physics and Signal Processing. Transferred to specialize in Data Science.

Certificate of Advanced English (C2) Cambridge Assessment *2019*

Projects

TripAdvisor Restaurants Data Pipeline *Python, Apache Airflow, PySpark, Kafka, FastAPI, React*

- Orchestrated a 5-task Airflow DAG (ETL + ML + streaming) over **1M+ rows** in 50k-row chunks: extract, clean, EDA, preprocess (normalization, OHE/label encoding by cardinality, incremental PCA), stream to Kafka.
- Applied PySpark ML KMeans clustering with automatic K selection via silhouette scoring (K = 10–80); OHE or label encoding per column based on cardinality; all parameters in `config.toml`.
- Built dual Spark Structured Streaming consumers: a fast filter path and an ML-scored path computing centroid distances and writing timestamped classifications in real time.
- Developed a full-stack web app (FastAPI + React/Vite): restaurant search via cosine similarity within clusters; live scoring tab pushing KMeans classifications to the browser via Server-Sent Events.

Forest Fire Spread Prediction *R, GLMs, GAMs, Statistical Modeling*

- Built GLM and GAM regression models to predict fire-burned area from meteorological covariates; GAMs outperformed GLMs on held-out data with lower residual variance.
- Resolved severe data quality issues (high skew, missing values, near-zero-variance predictors) via log transformation, imputation, and feature selection.
- Applied cross-validation to compare model fits; selected GAMs as the final model based on lower RMSE and better capture of non-linear covariate effects.

Stroke Risk Classification *Python, Scikit-learn, TensorFlow, ANN*

- Trained an MLP on an imbalanced medical dataset, applying SMOTE and optimizing the decision threshold to minimize false negatives given the clinical cost of missed strokes.
- Selected AUC-ROC and F1 as evaluation metrics over accuracy; tuned architecture and hyperparameters accordingly.
- Benchmarked the MLP against a logistic regression baseline; the non-linear model improved recall on the stroke class while maintaining acceptable precision.

Experience

Data Intelligence Platform Intern — Repsol Sep 2026 – Mar 2027

Talent Energy internship on the Data Intelligence Platform team, Madrid. Pipelines, orchestration and the infrastructure that keeps data moving.

Founder & President — Asociación Datalab May 2025 – Present

Established the university's first Data Science student association: defined the strategic roadmap, handled legal registration, and organized technical workshops.

Private Tutor — Mathematics & Physics Sep 2021 – Present

Mentored high-school students over 5 years, achieving a **100% pass rate** and helping 90% of students improve by at least 3 grade points through personalized plans.

Technical Skills

Programming: Python (Advanced), R, SQL, C
Data Science: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Hadoop, Spark, Matplotlib, Excel
Engineering: Apache Airflow, Kafka, Docker, Git/GitHub, Linux (Bash), Claude Code (agentic coding)
Languages: Spanish (Native), English (C2), German (B1), Chinese (HSK3)